Muhammad Hamza Naseem

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SOFTWARE ENGINEER

Results-driven Senior Software Engineer specializing in React, Angular, NodeJS and PostgreSQL. Built production LLM systems for a B2B procurement platform. Conversational AI agents, document intelligence, and semantic search across LangGraph, LangChain, and multi-provider model infrastructure (Google Vertex AI, Anthropic, OpenAI).

PROFESSIONAL EXPERIENCE

Senior Software Engineer, *Vroozi* 11/2023 – present
Lahore, Pakistan

- Built the document-information extraction pipeline that parses procurement documents (purchase requests, invoices) into structured data using Google Gemini on Vertex AI, driven by Pydantic output schemas and forced tool-calling for reliable structured output.Designed a config-driven, per-document-type model routing layer so each document type resolves its own model and inference parameters (thinking budget, reasoning effort, temperature) via a layered merge of app-level and per-request configuration, enabling model changes and A/B tuning with zero code deploys. Implemented a multi-provider chat-model factory abstracting Gemini (Vertex AI / GenAI), Anthropic Claude, Meta Llama (Vertex Model Garden), Mistral, and OpenAI behind one interface, with capability-gated params and multi-region Vertex AI fallbacks for resilience. Added truncated-response detection and automatic retry logic to harden extraction reliability against incomplete model outputs.
- Engineered a streaming, multi-turn conversational agent on LangGraph (StateGraph + ReAct-style agent/tool nodes with conditional routing) that lets users search catalog products, manage a shopping cart, and create purchase requests through natural language. Implemented token-level streaming via LangGraph astream_events, accumulating incremental text and deduplicated attachments for a responsive UI. Built human-in-the-loop tool execution using graph interrupt_before checkpoints, pausing the agent for client-side UI actions (cart updates, PR creation) and resuming with the result. Added persistent conversation memory / multi-turn state with a SQLite checkpointer keyed by session/thread, plus token-budgeted message trimming and recursion limits. Developed a self-corrective (CRAG-style) RAG subgraph: query contextualization -> retrieval -> document-relevance grading -> generation -> hallucination/groundedness grading -> answer-relevance grading, with a query-rewrite retry loop and cited references. Integrated reversible PII anonymization (Microsoft Presidio) - masking sensitive data before it reaches the model and restoring it in responses - and LangSmith tracing for observability.

- Built a FastAPI microservice that generates vector embeddings for catalog items and indexes them to power semantic/vector search across the procurement catalog. Designed pluggable embedding-provider and vector-store abstractions (factory/strategy pattern) supporting sentence-transformers (all-mpnet-base-v2, 768-dim) and OpenAI embeddings, backed by Elasticsearch dense_vector and MongoDB Atlas Vector Search with cosine similarity - all swappable via config.Implemented event-driven ingestion with async Kafka consumers (aiokafka) for vectorize/de-vectorize flows, publishing per-batch success/failure results back to Kafka. Optimized indexing with bulk upsert (Elasticsearch streaming_bulk, MongoDB bulk_write) and dynamic index mappings derived from embedding dimension and similarity metric. Added partial-success handling with per-item success/failure tracking, bounded retries with exponential backoff (tenacity), and New Relic observability.
- Integrated socket IO in Angular and configured its connection with Nodejs for real-time events.
- Developed and maintained multiple RESTful API endpoints using Node.js, ensuring efficient data handling and integration with front-end components.
- Worked on the migration of a large-scale application from AngularJS to Angular 9, modernizing the codebase and improving application performance.
- Conducted thorough E2E Testing along with writing unit tests to validate functionality and performance post-upgrade, ensuring the application met all quality standards.

Software Engineer, *Rolustech* 07/2021 – 11/2023
Lahore, Pakistan

- Implemented URL shortener microservice which generated a short URL for the provided URL using Angular, NodeJS and MongoDB.
- Deployed Angular and Node applications to AWS using AWS S3 and Elastic Beanstalk.
- Automated deployments on AWS through the implementation of CI/CD pipelines.
- Implemented internationalization and direction switching support across multiple portals.
- Utilized AWS CloudFront (CDN) to optimize application delivery speed.
- Integrated Stripe APIs into the backend server using NodeJS.
- Developed real-time event and notification systems using Pusher and Firebase Cloud Messaging on Angular and NodeJS.
- Established git pre-commit hooks with TSLint to enforce consistent coding standards for front-end and back-end.
- Implemented global scanning on a warehouse portal automating the tiresome work of 100+ warehouse workers with priority queue for various user flows.
- Synced warehouse inventory states in real-time for 100+ warehouse workers using IoT MQTT pub-sub model.
- Contributed to the implementation of flow diagrams and architectural decision designs, making 100+ use cases executable with a combination of less than 5 clicks.
- Conducted multiple sessions with a team of 15 developers on “The Pragmatic Programmer” book to adapt the practices of DRY, modularity, orthogonality, and API design by contracts, pre & post conditions.
- Integrated the Dexie indexedDB to cache the warehouse inventory on the client side, resulting in a 70% reduction in scan validation time.
- Scored 120+% task completion productivity throughout the project span and close to 200% in certain phases, decreasing feature release time.
- Leveraged AWS Lambda for serverless functions in NodeJS.

- Integrated Twilio's WhatsApp API using NodeJS for seamless messaging within the portal.
- Developed scalable APIs from scratch on NodeJS using Express, with efficient routing and middlewares.
- Significantly reduced application load time from 18 seconds to 12 seconds through modularization and lazy loading techniques.
- Implemented LRU caching on Angular and NodeJS for improved performance.
- Created metadata-based generic components to enhance code reusability.
- Applied the DRY principle to refactor and reduce codebase size by 10%.
- Implemented web workers to run CPU-intensive computations in a separate thread.
- Integrated Google Analytics for comprehensive website traffic tracking and reporting.
- Implemented service worker in Angular to listen for push API events from Firebase

EDUCATION

Bachelor in Computer Science, *Fast Nuces* 08/2017 – 07/2021
Lahore, Pakistan

Projects

Udacity
Made a short fullstack application like Udacity.com.
- **Technologies used:** NodeJS, React.

Image Detection System
Successfully trained a machine learning model to detect bounding boxes on Pakistan CNIC and Passport dataset using MASK-RCNN.
- **Technologies used:** Python, NumPy, Pandas, Machine Learning, and Matplotlib.

SKILLS

Programming Languages: TypeScript, JavaScript, Python
AI / LLM: LangGraph, LangChain, RAG, CRAG, Google Vertex AI (Gemini), Anthropic Claude, OpenAI, PyTorch, TensorFlow, Scikit-learn, XGBoost, sentence-transformers, Microsoft Presidio, LangSmith
Backend & Frameworks: FastAPI, Flask, Node.js, React, Angular, Async Programming, Pydantic, Kafka (aiokafka), Sequelize
Search, Data & Infrastructure: Elasticsearch Vector Search, MongoDB Atlas Vector Search, Semantic & Embedding Retrieval, MongoDB, PostgreSQL, Pandas, NumPy, Docker, AWS Lambda, AWS S3, Git, IndexedDB, LRU Cache